37. The name *two-to-two function* suggests that such a function f should carry every pair of distinct points into two distinct points. Such a function is one-to-one in the conventional sense. (If the domain has only one element, the function cannot fail to be two-to-two, because the only way it can fail to be two-to-two is to carry two points into one point, and the set does not have two points.) Conversely, every function that is one-to-one in the conventional sense carries each pair of distinct points into two distinct points. Thus the functions conventionally called one-to-one are precisely those that carry two points into two points, which is a much more intuitive unidirectional way of regarding them. Also, the standard way of trying to show that a function is one-to-one is precisely to show that it does not fail to be two-to-two. That is, proving that a function is one-to-one becomes more natural in the two-to-two terminology.

1. Introduction and Examples

1. $i^3 = i^2 \cdot i = -1 \cdot i = -i$ **2.** $i^4 = (i^2)^2 = (-1)^2 = 1$ **3.** $i^{23} = (i^2)^{11} \cdot i = (-1)^{11} \cdot i = (-1)i = -i$

4. $(-i)^{35} = (i^2)^{17}(-i) = (-1)^{17}(-i) = (-1)(-i) = i$

5. $(4-i)(5+3i) = 20 + 12i - 5i - 3i^2 = 20 + 7i + 3 = 23 + 7i$

6. $(8+2i)(3-i) = 24 - 8i + 6i - 2i^2 = 24 - 2i - 2(-1) = 26 - 2i$

7. $(2-3i)(4+i) + (6-5i) = 8 + 2i - 12i - 3i^2 + 6 - 5i = 14 - 15i - 3(-1) = 17 - 15i$

8. $(1+i)^3 = (1+i)^2(1+i) = (1+2i-1)(1+i) = 2i(1+i) = 2i^2 + 2i = -2 + 2i$

9. $(1-i)^5 = 1^5 + \frac{5}{1}1^4(-i) + \frac{5\cdot4}{2\cdot1}1^3(-i)^2 + \frac{5\cdot4}{2\cdot1}1^2(-i)^3 + \frac{5}{1}1^1(-i)^4 + (-i)^5 = 1 - 5i + 10i^2 - 10i^3 + 5i^4 - i^5 = 1 - 5i - 10 + 10i + 5 - i = -4 + 4i$

10. $|3-4i| = \sqrt{3^2 + (-4)^2} = \sqrt{9+16} = \sqrt{25} = 5$ 11. $|6+4i| = \sqrt{6^2 + 4^2} = \sqrt{36+16} = \sqrt{52} = 2\sqrt{13}$

12. $|3-4i| = \sqrt{3^2 + (-4)^2} = \sqrt{25} = 5$ and $3 - 4i = 5(\frac{3}{5} - \frac{4}{5}i)$

13. $|-1+i| = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$ and $-1 + i = \sqrt{2}(-\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i)$

14. $|12+5i| = \sqrt{12^2 + 5^2} = \sqrt{169}$ and $12 + 5i = 13(\frac{12}{13} + \frac{5}{13}i)$

15. $|-3+5i| = \sqrt{(-3)^2 + 5^2} = \sqrt{34}$ and $-3 + 5i = \sqrt{34}(-\frac{3}{\sqrt{34}} + \frac{5}{\sqrt{34}}i)$

16. $|z|^4(\cos 4\theta + i\sin 4\theta) = 1(1+0i)$ so $|z| = 1$ and $\cos 4\theta = 1$ and $\sin 4\theta = 0$. Thus $4\theta = 0 + n(2\pi)$ so $\theta = n\frac{\pi}{2}$ which yields values $0, \frac{\pi}{2}, \pi$, and $\frac{3\pi}{2}$ less than 2π. The solutions are

$$z_1 = \cos 0 + i\sin 0 = 1, \qquad z_2 = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} = i,$$

$$z_3 = \cos\pi + i\sin\pi = -1, \text{ and } z_4 = \cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2} = -i.$$

17. $|z|^4(\cos 4\theta + i\sin 4\theta) = 1(-1+0i)$ so $|z| = 1$ and $\cos 4\theta = -1$ and $\sin 4\theta = 0$. Thus $4\theta = \pi + n(2\pi)$ so $\theta = \frac{\pi}{4} + n\frac{\pi}{2}$ which yields values $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}$, and $\frac{7\pi}{4}$ less than 2π. The solutions are

$$z_1 = \cos\frac{\pi}{4} + i\sin\frac{\pi}{4} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i, \qquad z_2 = \cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4} = -\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i,$$

$$z_3 = \cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4} = -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i, \text{ and } z_4 = \cos\frac{7\pi}{4} + i\sin\frac{7\pi}{4} = \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i.$$

18. $|z|^3(\cos 3\theta + i\sin 3\theta) = 8(-1+0i)$ so $|z| = 2$ and $\cos 3\theta = -1$ and $\sin 3\theta = 0$. Thus $3\theta = \pi + n(2\pi)$ so $\theta = \frac{\pi}{3} + n\frac{2\pi}{3}$ which yields values $\frac{\pi}{3}, \pi$, and $\frac{5\pi}{3}$ less than 2π. The solutions are

$$z_1 = 2(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}) = 2(\frac{1}{2} + \frac{\sqrt{3}}{2}i) = 1+\sqrt{3}i, \qquad z_2 = 2(\cos\pi + i\sin\pi) = 2(-1+0i) = -2,$$

and

$$z_3 = 2(\cos\frac{5\pi}{3} + i\sin\frac{5\pi}{3}) = 2(\frac{1}{2} - \frac{\sqrt{3}}{2}i) = 1-\sqrt{3}i.$$

19. $|z|^3(\cos 3\theta + i\sin 3\theta) = 27(0-i)$ so $|z| = 3$ and $\cos 3\theta = 0$ and $\sin 3\theta = -1$. Thus $3\theta = 3\pi/2 + n(2\pi)$ so $\theta = \frac{\pi}{2} + n\frac{2\pi}{3}$ which yields values $\frac{\pi}{2}, \frac{7\pi}{6}$, and $\frac{11\pi}{6}$ less than 2π. The solutions are

$$z_1 = 3(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}) = 3(0+i) = 3i, \qquad z_2 = 3(\cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6}) = 3(-\frac{\sqrt{3}}{2} - \frac{1}{2}i) = -\frac{3\sqrt{3}}{2} - \frac{3}{2}i$$

and

$$z_3 = 3(\cos\frac{11\pi}{6} + i\sin\frac{11\pi}{6}) = 3(\frac{\sqrt{3}}{2} - \frac{1}{2}i) = \frac{3\sqrt{3}}{2} - \frac{3}{2}i.$$

20. $|z|^6(\cos 6\theta + i\sin 6\theta) = 1+0i$ so $|z| = 1$ and $\cos 6\theta = 1$ and $\sin 6\theta = 0$. Thus $6\theta = 0 + n(2\pi)$ so $\theta = 0 + n\frac{2\pi}{6}$ which yields values $0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}$, and $\frac{5\pi}{3}$ less than 2π. The solutions are

$$z_1 = 1(\cos 0 + i\sin 0) = 1+0i = 1, \qquad z_2 = 1(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}) = \frac{1}{2} + \frac{\sqrt{3}}{2}i,$$

$$z_3 = 1(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}) = -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \qquad z_4 = 1(\cos\pi + i\sin\pi) = -1+0i = -1,$$

$$z_5 = 1(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}) = -\frac{1}{2} - \frac{\sqrt{3}}{2}i, \qquad z_6 = 1(\cos\frac{5\pi}{3} + i\sin\frac{5\pi}{3}) = \frac{1}{2} - \frac{\sqrt{3}}{2}i.$$

21. $|z|^6(\cos 6\theta + i\sin 6\theta) = 64(-1+0i)$ so $|z| = 2$ and $\cos 6\theta = -1$ and $\sin 6\theta = 0$. Thus $6\theta = \pi + n(2\pi)$ so $\theta = \frac{\pi}{6} + n\frac{2\pi}{6}$ which yields values $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{3\pi}{2}$ and $\frac{11\pi}{6}$ less than 2π. The solutions are

$$\begin{aligned}
z_1 &= 2(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}) = 2(\frac{\sqrt{3}}{2} + \frac{1}{2}i) = \sqrt{3}+i,\\
z_2 &= 2(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}) = 2(0+i) = 2i,\\
z_3 &= 2(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}) = 2(-\frac{\sqrt{3}}{2} + \frac{1}{2}i) = -\sqrt{3}+i,\\
z_4 &= 2(\cos\frac{7\pi}{6} + i\sin\frac{7\pi}{6}) = 2(-\frac{\sqrt{3}}{2} - \frac{1}{2}i) = -\sqrt{3}-i,\\
z_5 &= 2(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}) = 2(0-i) = -2i,\\
z_6 &= 2(\cos\frac{11\pi}{6} + i\sin\frac{11\pi}{6}) = 2(\frac{\sqrt{3}}{2} - \frac{1}{2}i) = \sqrt{3}-i.
\end{aligned}$$

22. $10+16 = 26 > 17$, so $10 +_{17} 16 = 26 - 17 = 9$.

23. $8+6 = 14 > 10$, so $8 +_{10} 6 = 14 - 10 = 4$.

24. $20.5 + 19.3 = 39.8 > 25$, so $20.5 +_{25} 19.3 = 39.8 - 25 = 14.8$.

25. $\frac{1}{2} + \frac{7}{8} = \frac{11}{8} > 1$, so $\frac{1}{2} +_1 \frac{7}{8} = \frac{11}{8} - 1 = \frac{3}{8}$.

26. $\frac{3\pi}{4} + \frac{3\pi}{2} = \frac{9\pi}{4} > 2\pi$, so $\frac{3\pi}{4} +_{2\pi} \frac{3\pi}{2} = \frac{9\pi}{4} - 2\pi = \frac{\pi}{4}$.

27. $2\sqrt{2}+3\sqrt{2}=5\sqrt{2}>\sqrt{32}=4\sqrt{2}$, so $2\sqrt{2}+_{\sqrt{32}}\ 3\sqrt{2}=5\sqrt{2}-4\sqrt{2}=\sqrt{2}$.

28. 8 is not in $\mathbb{R}_6$ because $8>6$, and we have only defined $a+_6 b$ for $a,b\in\mathbb{R}_6$.

29. We need to have $x+7=15+3$, so $x=11$ will work. It is easily checked that there is no other solution.

30. We need to have $x+\frac{3\pi}{2}=2\pi+\frac{3\pi}{4}=\frac{11\pi}{4}$, so $x=\frac{5\pi}{4}$ will work. It is easy to see there is no other solution.

31. We need to have $x+x=7+3=10$, so $x=5$ will work. It is easy to see that there is no other solution.

32. We need to have $x+x+x=7+5$, so $x=4$ will work. Checking the other possibilities 0, 1, 2, 3, 5, and 6, we see that this is the only solution.

33. An obvious solution is $x=1$. Otherwise, we need to have $x+x=12+2$, so $x=7$ will work also. Checking the other ten elements, in $\mathbb{Z}_{12}$, we see that these are the only solutions.

34. Checking the elements $0,1,2,3\in\mathbb{Z}_4$, we find that they are all solutions. For example, $3+_4 3+_4 3+_4 3=(3+_4 3)+_4(3+_4 3)=2+_4 2=0$.

35. $\zeta^0\leftrightarrow 0,\quad \zeta^3=\zeta^2\zeta\leftrightarrow 2+_8 5=7,\quad \zeta^4=\zeta^2\zeta^2\leftrightarrow 2+_8 2=4,\quad \zeta^5=\zeta^4\zeta\leftrightarrow 4+_8 5=1,$
$\zeta^6=\zeta^3\zeta^3\leftrightarrow 7+_8 7=6,\quad \zeta^7=\zeta^3\zeta^4\leftrightarrow 7+_8 4=3$

36. $\zeta^0\leftrightarrow 0,\quad \zeta^2=\zeta\zeta\leftrightarrow 4+_7 4=1,\quad \zeta^3=\zeta^2\zeta\leftrightarrow 1+_7 4=5,\quad \zeta^4=\zeta^2\zeta^2\leftrightarrow 1+_7 1=2,$
$\zeta^5=\zeta^3\zeta^2\leftrightarrow 5+_7 1=6,\quad \zeta^6=\zeta^3\zeta^3\leftrightarrow 5+_7 5=3$

37. If there were an isomorphism such that $\zeta\leftrightarrow 4$, then we would have $\zeta^2\leftrightarrow 4+_6 4=2$ and $\zeta^4=\zeta^2\zeta^2\leftrightarrow 2+_6 2=4$ again, contradicting the fact that an isomorphism $\leftrightarrow$ must give a *one-to-one correpondence*.

38. By Euler's fomula, $e^{ia}e^{ib}=e^{i(a+b)}=\cos(a+b)+i\sin(a+b)$. Also by Euler's formula,

$$\begin{aligned} e^{ia}e^{ib} &= (\cos a+i\sin a)(\cos b+i\sin b)\\ &= (\cos a\cos b-\sin a\sin b)+i(\sin a\cos b+\cos a\sin b). \end{aligned}$$

The desired formulas follow at once.

39. (See the text answer.)

40. **a.** We have $e^{3\theta}=\cos 3\theta+i\sin 3\theta$. On the other hand,

$$\begin{aligned} e^{3\theta} &= (e^{\theta})^3=(\cos\theta+i\sin\theta)^3\\ &= \cos^3\theta+3i\cos^2\theta\sin\theta-3\cos\theta\sin^2\theta-i\sin^3\theta\\ &= (\cos^3\theta-3\cos\theta\sin^2\theta)+i(3\cos^2\theta\sin\theta-\sin^3\theta). \end{aligned}$$

Comparing these two expressions, we see that

$$\cos 3\theta=\cos^3\theta-3\cos\theta\sin^2\theta.$$

b. From Part(**a**), we obtain

$$\cos 3\theta=\cos^3\theta-3(\cos\theta)(1-\cos^2\theta)=4\cos^3\theta-3\cos\theta.$$